

Class 20 Pythagoras and Diophantus

Goals

1. State and prove the Pythagorean Theorem.
2. Know the definition of a Diophantine equation.
3. Solve linear Diophantine equations.

Review: Linear combination theorem:

What does this say if a and b are relatively prime?

What does Euclid's Lemma say?

Definition: A Diophantine equation is a polynomial equation in at least two variables with integer solutions.

Which of the following are Diophantine equations?

1. $3x + 4y = 1$

2. $3x + 4y = 10$

3. $3x + 4y = c \in \mathbb{Z}$

4. $14x + 35y = 1$

These are examples of linear Diophantine equations.

Question: For what values of c will $ax+by = c$ have a solution?

Perhaps the most famous Diophantine equation arises from a geometric fact about right triangles known for thousands of years.

If a , b , and c represent the lengths of the sides of any right triangle with c being the length of the longest side, then

$$a^2 + b^2 = c^2.$$

The Diophantine question asks if there are right triangles with integer length sides.

Are there positive integers that satisfy the equation

$$x^2 + y^2 = z^2$$

Definition: A Pythagorean triple is

Examples

Prove that (12, 35, 37) is a Pythagorean triple.

Prove: If n is any odd number greater than 1, then $\left(n, \frac{n^2 - 1}{2}, \frac{n^2 + 1}{2}\right)$ is a Pythagorean triple.

Question: Does $xy - 5x + 6y = 0$ have any integer solutions?

Does it have any positive integer solutions?

Try $3xy - 11x + 5y = 0$